

JIAWEI CHEN

Crystal Structure Prediction (CSP) / Machine Learning (ML) / First-Principles Calculations

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 Guangdong University of Technology  M.S. in Materials Physics and Chemistry

Education

2027.06	School of Physics and Optoelectronic Engineering, Guangdong University of Technology
2024.09	M.S. Candidate GPA: 4.01/5.0 in Materials Physics and Chemistry
2024.06	School of Chemical Engineering and Light Industry, Guangdong University of Technology
2020.09	B.S. GPA: 3.73/5.0 in Applied Chemistry

Honors & Awards

- › Exempted from Graduate Entrance Examination
- › Graduate Elite Talent Program, Guangdong University of Technology
- › First-Class Graduate Scholarship (2024, 2024–2025), Guangdong University of Technology
- › Third Prize, South China Undergraduate Physics Experiment Competition (2022)
- › Second Prize, Guangdong Division of the National Mathematical Contest in Modeling (2022)
- › Outstanding Student Scholarship (2020–2023), Guangdong University of Technology

Research Experience

2026.05	USPEX-Analyzer: Browser-based analysis tool for USPEX V10.5
2026.04	› Developed a browser-based post-processing tool for USPEX outputs, supporting interactive convex-hull/Pareto/genealogy visualization, multi-condition filtering, and export
Present	CSP of Hydrogen-Based Superconductors via Multi-Objective Optimization
2026.01	› Embedded Interpretable Prediction Model into USPEX's multi-objective optimization to co-optimize thermodynamic stability and superconducting properties
	› Establishing a closed-loop workflow of search–screen–validate–update
Present	CSP Driven by Universal Machine Learning Potentials
2025.01	› Integrated universal ML potentials into USPEX for faster CSP
	› Fine-tuned MatterSim to improve prediction accuracy under high pressure
2026.01	High-Throughput Screening of Conventional Superconductors
2025.01	› Combined structure generators with MatterSim for high-throughput screening of hydrogen-based superconductors
	› Accelerated substitution screening of layered boride superconductors using MatterSim
2026.01	Interpretable Prediction Model for T_c of Hydrogen-Based Superconductors
2024.09	› Collected, preprocessed datasets and performed feature engineering
	› Constructed interpretable models using SISSO
	› Revealed relationships between pressure, electronic structure, and superconductivity
2024	Defect Engineering for Hydrogen Evolution Reaction in 2D Materials
2022	› Investigated the effects of defects on electronic structure and HER activity of HfX_2 ($X = \text{S}, \text{Se}, \text{Te}$)

Skills

Programming	Python, Bash, LaTeX
Software	VASP, Quantum ESPRESSO, USPEX, SISSO, Lammmps

Publications

- › **Chen J**, Peng J, Liang Y, et al. Interpretable descriptors enable prediction of hydrogen-based superconductors at moderate pressures. *Materials Today Physics*, 2026, 63: 102073.
- › Liang Y*, **Chen J***, Huang Y, et al. High-Throughput Screening of Bulk Boride Superconductors with Honeycomb-Ruby Frameworks. *Inorganic Chemistry*, 2026, 65(10): 5731–5739.
- › **Chen J***, Zhang R*, Luo J, et al. Activating HfX_2 ($X = \text{S}, \text{Se}$ and Te) for the hydrogen evolution reaction by introducing defects: a first-principles study. *Physical Chemistry Chemical Physics*, 2023, 25(38): 26043–26048.